Complete papers and shared theories

Thirteen measured sets, thirty-three original manuscripts

This additive overview brings together admitted values. It preserves the earlier 27-original report, adds Matching, Chiral enumeration and the N007 comparison, and uses the enlarged A, B and C editions. The numerical baseline is commit `c6b1b13`. The subsequent N006 admission is shown separately as *unmeasured*; the old N007 factor does not describe the enlarged orbit-construction topic.

For each topic, $S = \sum_i c(L_i)$ counts the complete originals, U every complete common paper, and $A = \sum_i c(A_i)$ every complete application. The two headline comparisons are

$$R = \frac{S}{U + A}, \qquad r_i = \frac{c(L_i)}{c(A_i)} \quad \text{after acquiring all common theory and declared programs.}$$

Every topic pays its common theory independently, even when proofs overlap other topics. A factor above one means a shorter replacement in the stated coordinate; a factor below one means a longer replacement. Conditional factors are not standalone application savings.

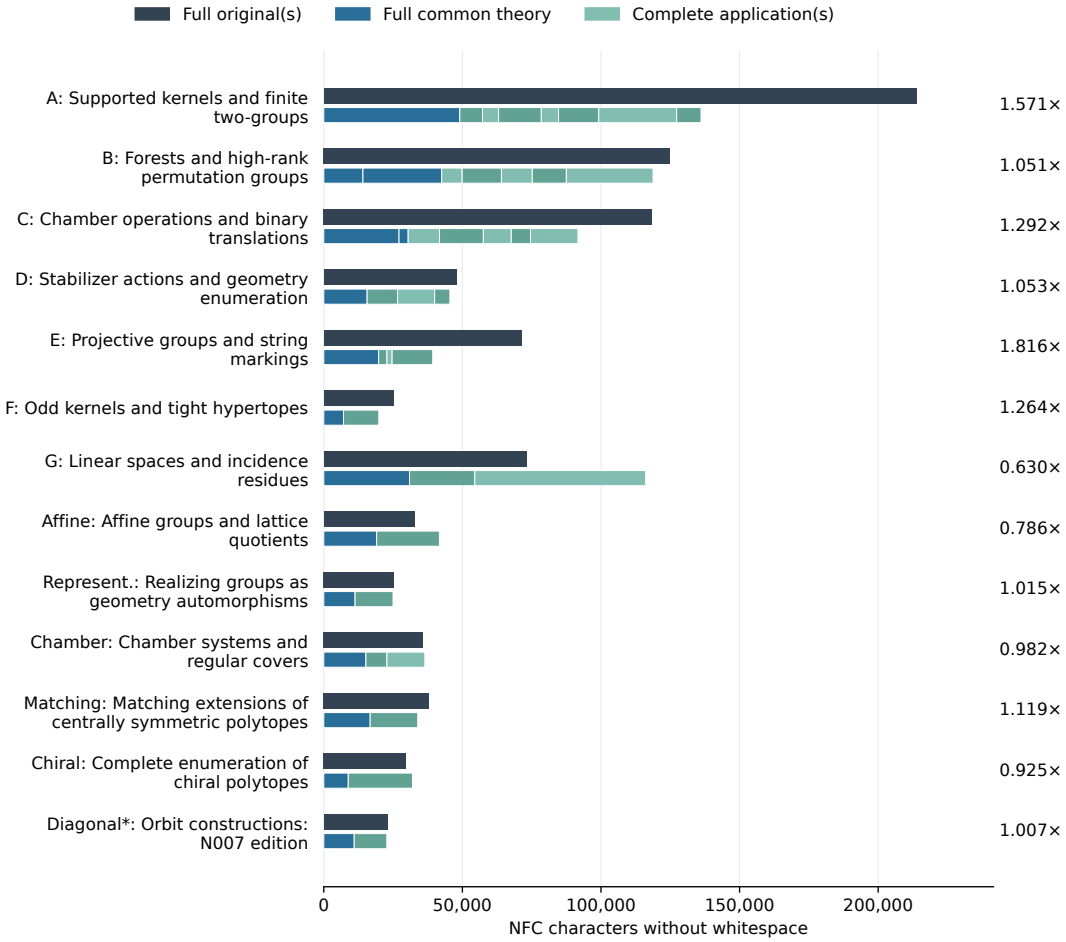
Complete written costs. All values below are NFC characters excluding whitespace. The fixed whole-document boundaries include their introductions, proofs, local context and references.

Topic	Originals	Sources	Common	Applications	Written	Factor
A	7	213,803	49,059	87,019	136,078	1.571
B	5	124,906	42,547	76,243	118,790	1.051
C	5	118,433	30,437	61,248	91,685	1.292
D	3	47,894	15,632	29,855	45,487	1.053
E	3	71,375	19,790	19,513	39,303	1.816
F	1	25,115	7,118	12,744	19,862	1.264
G	2	73,199	30,928	85,207	116,135	0.630
Affine	1	32,767	19,070	22,638	41,708	0.786
Represent.	1	25,348	11,288	13,688	24,976	1.015
Chamber	2	35,802	15,199	21,254	36,453	0.982
Matching	1	37,951	16,732	17,168	33,900	1.119
Chiral	1	29,599	8,831	23,171	32,002	0.925
Diagonal*	1	22,888	10,968	11,771	22,739	1.007

Representations remain distinct. B and C use complete authored text renderings for N001 and L108; all other source and replacement PDFs in those two topics are counted raw and untrimmed. Their older original-PDF coordinates remain unknown. The other rows use accepted calibrated, selected PDF text with exact repeated furniture removed. No pooled factor across these representations is formed.

Scope. These are 33 measured originals within the fixed 161-item programme. The current scientific register has 34 admitted recoveries among 104 accessible originals, with N006 awaiting measurement; 70 accessible originals await complete recovery and 57 remain retrieval targets. Complete numerators match the declared principal-result contracts, with corrections, retained facts and ancillary exclusions preserved. This report changes no scientific, question or formal credit.

Full common theories and applications. Each row has one scale for its original sum and replacement. Blue segments are whole common papers; green segments are individual complete applications. B pays both common papers (14,124 and 28,423). C pays its original common paper (27,142) and the entire incidence note (3,295). All segments remain charged.



The Diagonal row concerns only the earlier N007 edition. There is no bar or numerical placeholder for N006, and no measured whole-topic factor for the expanded N006+N007 edition. Text length is not a minimum proof complexity or a measure of mathematical originality.

Complete programs and prior acquisition. The written comparison remains first. These separate alternatives add the entire declared printed appendix, or its entire raw program files, to the written denominator. Printed and raw code are never added together.

Topic	Printed code	With printed	Raw code	With raw	U+printed
A	54,210	1.124	52,807	1.132	103,269
B	13,170	0.947	12,330	0.953	55,717
C	20,933	1.052	19,937	1.061	51,370
D	27,070	0.660	26,185	0.668	42,702
E	7,565	1.523	–	–	27,355
F	8,837	0.875	8,061	0.899	15,955
G	53,018	0.433	50,184	0.440	83,946
Affine	17,250	0.556	16,682	0.561	36,320
Represent.	9,025	0.746	8,310	0.762	20,313
Chamber	7,677	0.811	6,401	0.835	22,876
Matching	4,998	0.976	4,408	0.991	21,730
Chiral	11,940	0.674	11,090	0.687	20,771
Diagonal*	3,161	0.884	2,856	0.894	14,129

The final column is prepaid in every conditional row for that topic. E preserves its admitted convention: its optional validation program costs another 5,173 characters and gives a separate program-inclusive factor of 1.372. An unadmitted raw-code alternative for E is not invented here.

Program roles. A includes all five files. B includes one finite proof premise, two validations and an auxiliary example. C includes its three complete GAP files; the identical shared construction is charged once, and its exhaustive census remains paid. D includes five required and three diagnostic programs. E separates one required program from optional validation. F's three programs and Representation's two are validations, not uniform proof premises. G retains four finite computations, two certificate bridges and seven validations. Affine retains all six mathematical programs. Chamber retains one required GAP program and one diagnostic. Matching's census is a finite premise. Chiral enumeration retains all three enumeration/formula/validation programs. The N007 edition includes its whole validation program. Separate operational drivers and partial Lean inventories stay separate, as in the admitted packets.

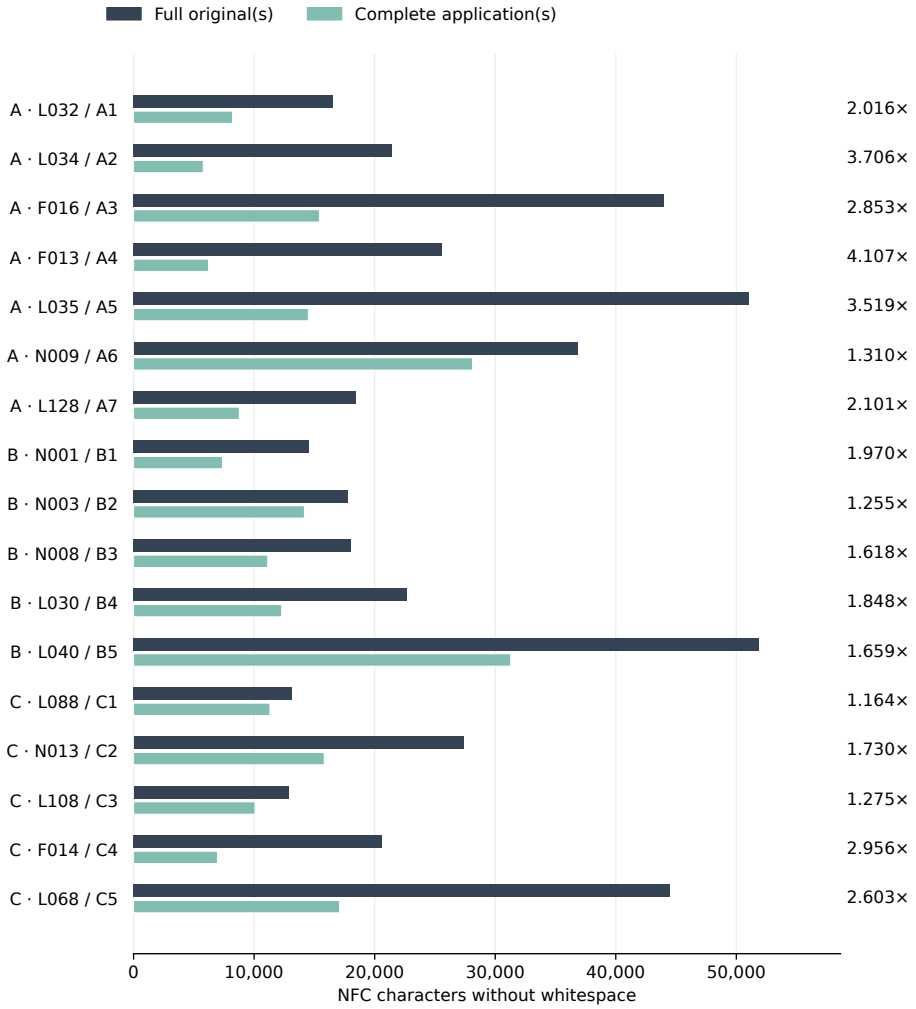
Limits. Retained full external proofs, unprovided historical software, complete graphic information and end-to-end formalization have unknown costs, not zero. Units matter: Representation's written factor is about 0.971 in its admitted UTF-8 coordinate, and Chiral enumeration is about 1.003 in whitespace tokens but 0.925 in nonspace characters. All underlying multi-unit sensitivities remain available; this table consistently uses the declared primary character coordinate.

Every complete application after full acquisition. All common papers and all declared programs in the preceding table have already been paid. The source and application are complete documents under the same topic boundary; no smaller theorem closure replaces either side.

Topic	Original	Application	Source cost	Application cost	Conditional
A	L032	A1	16,558	8,212	2.016
A	L034	A2	21,433	5,784	3.706
A	F016	A3	43,983	15,419	2.853
A	F013	A4	25,562	6,224	4.107
A	L035	A5	51,019	14,498	3.519
A	N009	A6	36,824	28,114	1.310
A	L128	A7	18,424	8,768	2.101
B	N001	B1	14,519	7,371	1.970
B	N003	B2	17,780	14,170	1.255
B	N008	B3	18,009	11,132	1.618
B	L030	B4	22,704	12,286	1.848
B	L040	B5	51,894	31,284	1.659
C	L088	C1	13,166	11,311	1.164
C	N013	C2	27,361	15,815	1.730
C	L108	C3	12,849	10,075	1.275
C	F014	C4	20,577	6,962	2.956
C	L068	C5	44,480	17,085	2.603
D	L066	D1	20,916	10,989	1.903
D	L074	D2	20,661	13,367	1.546
D	L104	D3	6,317	5,499	1.149
E	L083	E1	20,885	3,002	6.957
E	L079	E2	20,908	1,855	11.271
E	L021	E3	29,582	14,656	2.018
F	L014	F1	25,115	12,744	1.971
G	L009	G1	49,891	23,544	2.119
G	L092	G2	23,308	61,663	0.378
Affine	L012	AF1	32,767	22,638	1.447
Represent.	F010	R1	25,348	13,688	1.852
Chamber	L008	H1	21,784	7,571	2.877
Chamber	N016	H2	14,018	13,683	1.024
Matching	F015	M1	37,951	17,168	2.211
Chiral	L028	CE1	29,599	23,171	1.277
Diagonal*	N007	DG1	22,888	11,771	1.944

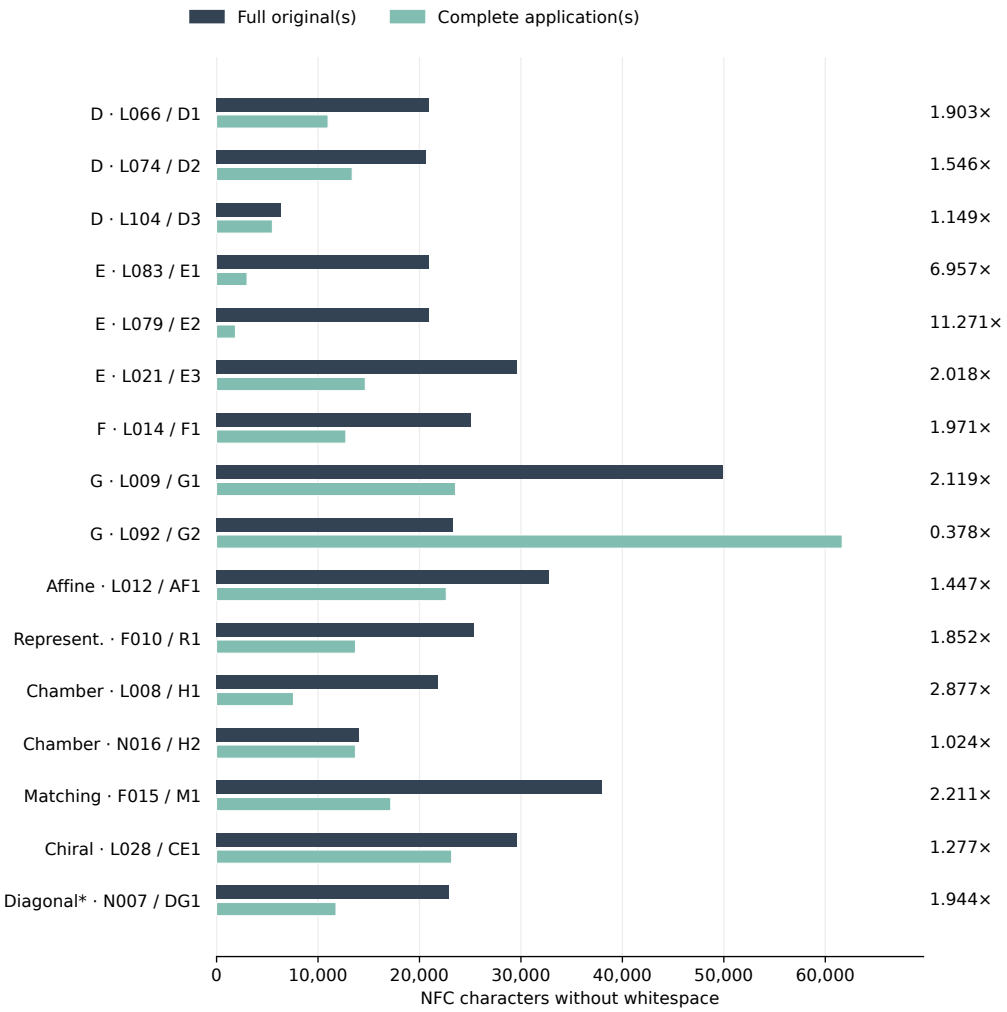
The asterisk on Diagonal identifies the earlier N007-only acquisition. These individual factors cannot be transferred to a new common edition without a newly fixed measurement. No node-count factor is used.

Individual source and application costs: first seventeen originals. Dark bars show complete originals; green bars show their complete applications. Factors at right assume full prior acquisition, including the enlarged B and C common theory and all declared programs. A shared scale is used within this chart.



Source IDs are stable catalogue labels. Their exact definitions, source editions and proof contracts remain in the linked admitted measurements; they are not additional mathematical claims of this overview.

Individual source and application costs: remaining sixteen originals. The same conventions apply; the scale is shared by every source/application pair in this chart. No cost for N006 is inferred from the last row.



Exact data and provenance. The accompanying `results.json`, `topics.csv` and `applications.csv` retain exact integer costs and rational factors. `protocol.json` pins the old admitted overview, four admitted numerical updates, their independent records and the unmeasured N006 admission. No primary text was copied or extracted in this consolidation. Graph bars use these same costs; they do not depict theorem counts or claims of optimality. The new consolidation requires distinct review before publication.